

Predefining an analysis plan

Project: Reproducible AI Research Study

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Privacy and data governance

Without opening the old lesson, explain one key idea from Privacy and data governance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Dataset and benchmark limitations

Without opening the old lesson, explain one key idea from Dataset and benchmark limitations. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Multiple comparisons

Without opening the old lesson, explain one key idea from Multiple comparisons. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Predefining an analysis plan. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Predefining an analysis plan. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Plan excludes malformed inputs by a rule fixed beforehand. Three of 15 are malformed; 9 of 12 eligible pass. Report.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Plan includes all 20 cases. After results, the team removes four failed cases because they make the graph look worse. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A prespecified plan can still be flawed; transparency prevents hidden changes, not invalid assumptions. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Plan excludes malformed inputs by a rule fixed beforehand. Three of 15 are malformed; 9 of 12 eligible pass. Report.

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

Add a labelled predefining an analysis plan evidence sheet to Reproducible AI Research Study. Show how you can explain the mechanism and terminology of predefining an analysis plan. Use the supplied fictional case and keep its inputs, intermediate steps, calculation or prediction, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Predefining an analysis plan

Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.

Plan: compare correct counts on 12 cases; exclude only unreadable files identified before scoring; report all version results.

Predefining an analysis plan

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Write the rule first

Record the outcome and justified unreadable-file exclusion.



2. Protect the reserved set

Do not choose exclusions from which cases each model fails.



3. Report deviations

If the plan changes, mark the change and give its reason.

CHANGE ONE THING

Exclude two readable cases because the new model fails them.

Explain the changed result using the matching lesson.

Apply the recorded exclusion rule. Can two readable model failures be removed, and how must a necessary later plan change be disclosed?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

13. Project portfolio milestone

Project: Reproducible AI Research Study

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Freeze decisions before the result. A team records its outcome, exclusions and analysis before evaluating a reserved set.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? If the plan changes, mark the change and give its reason.

1. Protect the reserved set
2. Write the rule first
3. Report deviations

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Report deviations If the plan changes, mark the change and give its reason. An analysis plan limits flexibility after outcomes are known. Necessary changes can be transparent, but they should not be silently presented as the original plan.

Practice 2: explain the deciding evidence

Changed-case practice: Exclude two readable cases because the new model fails them. Which conclusion is supported by the supplied facts?

1. That outcome-driven exclusion violates the recorded rule.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: That outcome-driven exclusion violates the recorded rule. Model failure is not the predeclared unreadable-file criterion.

Repair a missing building block

Review Privacy and data governance (ai-7/20). Goal: Explain the mechanism and terminology of privacy and data governance. Short practice: Point to the analysis table and the restricted identity key. Why are the codes pseudonymous rather than guaranteed anonymous, and what changes if the key is published?. Return to this week and explain what you changed.

Review Dataset and benchmark limitations (ai-7/18). Goal: Explain the mechanism and terminology of dataset and benchmark limitations. Short practice: Report the indoor benchmark with its denominator and missing condition. After two correct outdoor cases out of four are added, what slice rate is supported and what remains uncertain?. Return to this week and explain what you changed.

Review Multiple comparisons (ai-7/15). Goal: Explain the mechanism and terminology of multiple comparisons. Short practice: Calculate the six-test cutoff and identify which supplied p-values reject. Why must the team count every planned comparison rather than report only its smallest p-value?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

